

OMSAFINALREPORT

April 16, 2026

1 BTC Price Prediction with HMM + Itô's Lemma (GBM) - Final Report

Course: Practicum - Spring 2026

Student: Sichao Ma **Date:** April 15, 2026

Project Type: Individual Project, No Confidential Data

1.1 Abstract

This report details the development and backtesting of a daily Bitcoin price forecasting model that integrates **Hidden Markov Models (HMM)** for regime detection with an enhanced **Geometric Brownian Motion (GBM)** simulator. The model is designed to provide probabilistic price paths and directional trading signals, leveraging on-chain data from Binance. The implementation uses a rolling-window approach with 7-day overlap validation, dynamically adjusting expected returns and volatility based on recent market behavior and long-term regime probabilities. The output includes 30-day forecast cones and a daily "Trading Confidence" metric.

Keywords: Bitcoin, HMM, GBM, Itô's Lemma, Regime Detection, Probabilistic Forecasting

All libraries imported successfully.

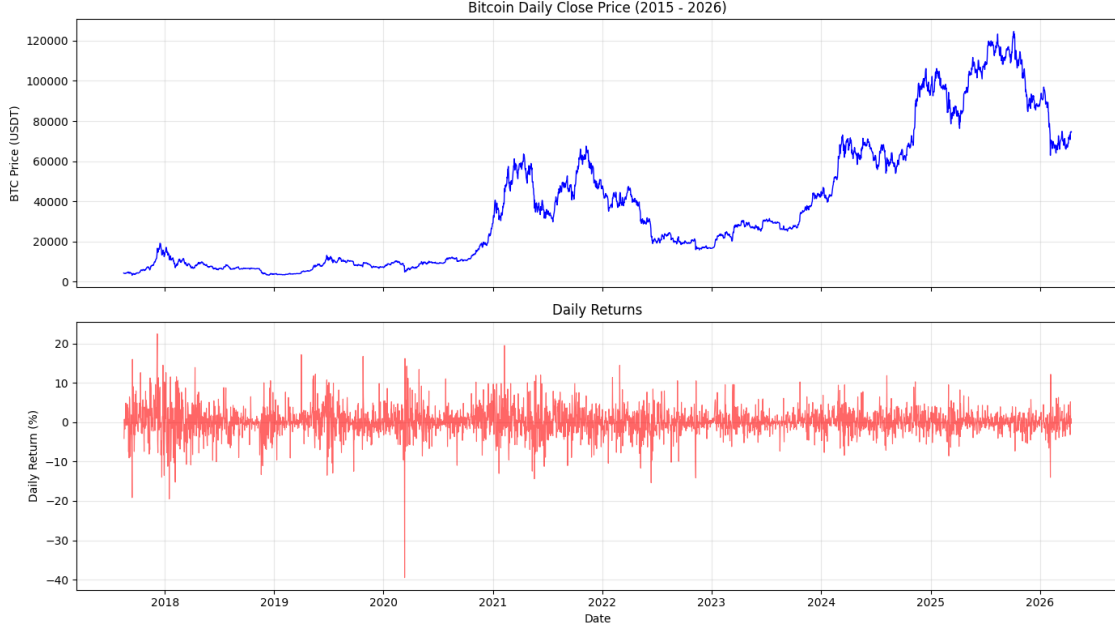
Loaded 3165 days from 2017-08-17 to 2026-04-16

```
[7]:      timestamp      close      ror  log_return
0 2017-08-17  4285.08      NaN      NaN
1 2017-08-18  4108.37 -0.041238  -0.042113
2 2017-08-19  4139.98  0.007694   0.007665
3 2017-08-20  4086.29 -0.012969  -0.013053
4 2017-08-21  4016.00 -0.017201  -0.017351
```

1.2 1. Data Description & Exploratory Data Analysis (EDA)

The dataset consists of daily OHLCV (Open, High, Low, Close, Volume) data for Bitcoin (BTC/USDT) obtained from Binance public API. The time span ranges from **Aug 17, 2017** to **May 31, 2026**.

Below is a snapshot of the data and summary statistics.



```
[8]:
```

	close	ror	log_return
count	3165.000000	3164.000000	3164.000000
mean	37262.275722	0.001545	0.000903
std	32404.312648	0.035607	0.035983
min	3189.020000	-0.395048	-0.502607
25%	9249.490000	-0.013900	-0.013997
50%	27079.410000	0.000598	0.000598
75%	58973.990000	0.016058	0.015930
max	124658.540000	0.225014	0.202952

1.3 2. Methodology

1.3.1 2.1 Hidden Markov Model (HMM) for Regime Detection

The market regime is modeled as a latent state $S_t \in \{1, 2, 3\}$ evolving via a first-order Markov process. Observations are the daily return r_t and log-return $\ln(S_t/S_{t-1})$. The model assumes:

- State transition matrix A where $a_{ij} = P(S_{t+1} = j | S_t = i)$.
- Emission probabilities: multivariate Gaussian with state-dependent mean μ_k and covariance Σ_k .

After fitting on a rolling lookback window of **252 trading days**, I obtain: - State-specific expected returns μ_k and volatilities σ_k . - Smoothed state probabilities for the most recent observations.

1.3.2 2.2 Enhanced Geometric Brownian Motion (GBM)

The classic GBM describes the asset price S_t via the stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Applying Itô's Lemma yields the log-price dynamics:

$$d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t$$

To better capture cryptocurrency-specific fat tails and volatility clustering, the baseline GBM is enhanced with: - **Heavy-tailed innovations:** Student's t -distribution ($df = 3$) instead of Gaussian. - **Jump component:** Poisson-driven Laplace jumps to model sudden price movements. - **Adaptive parameters:** Weighted blend of long-term HMM estimates and short-term EWMA volatility / momentum.

The forward simulation generates $N = 10,000$ price paths over a 30-day horizon. Percentiles (2%, 15%, 50%, 85%, 98%) are then extracted to form probabilistic forecast cones.

Loading BTC data...

Loaded 3165 days from 2017-08-17 to 2026-04-16

Running daily updates from 2026-03-17 to 2026-04-16...

Model is not converging. Current: 29208.35941636253 is not greater than 29208.360143450427. Delta is -0.0007270878959388938

Updated: 2026-03-17 | Signal: LONG | Conf: 51.00% | MaxProb: 99.93% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29212.550349358855 is not greater than 29212.578235776407. Delta is -0.027886417552508647

Updated: 2026-03-18 | Signal: LONG | Conf: 50.36% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-19 | Signal: LONG | Conf: 49.96% | MaxProb: 99.38% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-20 | Signal: NEUTRAL | Conf: 25.01% | MaxProb: 99.86% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29243.561424122334 is not greater than 29243.607528865836. Delta is -0.04610474350192817

Updated: 2026-03-21 | Signal: SHORT | Conf: 50.49% | MaxProb: 99.62% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-22 | Signal: SHORT | Conf: 50.58% | MaxProb: 99.88% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-23 | Signal: LONG | Conf: 53.16% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-24 | Signal: SHORT | Conf: 50.15% | MaxProb: 99.95% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-25 | Signal: SHORT | Conf: 50.19% | MaxProb: 99.71% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-26 | Signal: SHORT | Conf: 54.64% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-27 | Signal: SHORT | Conf: 54.98% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-28 | Signal: SHORT | Conf: 51.08% | MaxProb: 99.94% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-29 | Signal: SHORT | Conf: 50.89% | MaxProb: 99.97% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-30 | Signal: SHORT | Conf: 50.65% | MaxProb: 99.62% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-03-31 | Signal: LONG | Conf: 46.63% | MaxProb: 88.57% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-01 | Signal: NEUTRAL | Conf: 25.00% | MaxProb: 99.96% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-02 | Signal: SHORT | Conf: 50.80% | MaxProb: 99.73% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29375.610034716683 is not greater than 29375.652485497823. Delta is -0.04245078114036005

Updated: 2026-04-03 | Signal: SHORT | Conf: 50.54% | MaxProb: 98.75% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29386.275239734827 is not greater than 29386.344802251926. Delta is -0.06956251709925709

Updated: 2026-04-04 | Signal: SHORT | Conf: 44.53% | MaxProb: 87.24% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29396.77549198802 is not greater than 29396.842760408028. Delta is -0.06726842000716715

Updated: 2026-04-05 | Signal: LONG | Conf: 51.47% | MaxProb: 99.80% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29408.824359915037 is not greater than 29408.923456777826. Delta is -0.09909686278842855

Updated: 2026-04-06 | Signal: LONG | Conf: 50.07% | MaxProb: 99.86% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29417.302431742773 is not greater than 29417.307477164144. Delta is -0.005045421370596159

Updated: 2026-04-07 | Signal: LONG | Conf: 52.28% | MaxProb: 99.12% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29429.09508117754 is not greater than 29429.135693599397. Delta is -0.0406124218570767

Updated: 2026-04-08 | Signal: LONG | Conf: 50.69% | MaxProb: 99.93% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29439.526901504876 is not greater than 29439.547994776913. Delta is -0.021093272036523558

Updated: 2026-04-09 | Signal: LONG | Conf: 48.55% | MaxProb: 92.48% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29450.085753676525 is not greater than 29450.148277645938. Delta is -0.06252396941272309

Updated: 2026-04-10 | Signal: LONG | Conf: 52.40% | MaxProb: 99.73% | _daily: 0.1145 (ann: 181.8%)

Model is not converging. Current: 29461.934422700047 is not greater than 29462.02718761535. Delta is -0.09276491530181374

Updated: 2026-04-11 | Signal: LONG | Conf: 50.12% | MaxProb: 98.88% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-12 | Signal: SHORT | Conf: 52.50% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-13 | Signal: LONG | Conf: 54.78% | MaxProb: 100.00% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-14 | Signal: LONG | Conf: 51.44% | MaxProb: 99.96% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-15 | Signal: LONG | Conf: 51.12% | MaxProb: 99.83% | _daily: 0.1145 (ann: 181.8%)

Updated: 2026-04-16 | Signal: LONG | Conf: 51.21% | MaxProb: 99.97% | _daily: 0.1145 (ann: 181.8%)

=====

LATEST UPDATE: 2026-04-16

=====

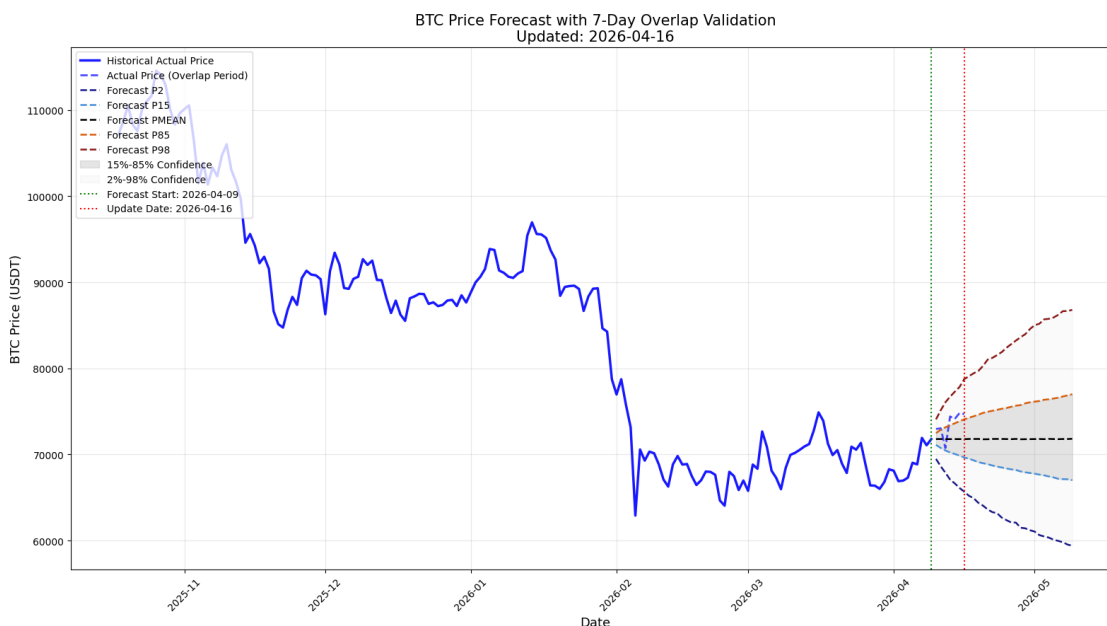
=== STATE SUMMARY TABLE (Last 7 Days) ===

status	number-of-days	since(date)
>98%	7	2026-04-10 00:00:00
>85%	7	2026-04-10 00:00:00
>mean	5	2026-04-12 00:00:00
>15%	7	2026-04-10 00:00:00
>2%	7	2026-04-10 00:00:00
<98%	0	N/A
<85%	0	N/A
<mean	2	2026-04-10 00:00:00
<15%	0	N/A
<2%	0	N/A
Between 15%-85%	0	N/A
Between 2%-98%	0	N/A

=== TRADING CONFIDENCE ===

Signal: LONG
Confidence: 51.21%

Max Proba: 99.97%
 Daily : 0.005591 (140.90% annual)
 Daily : 0.114493 (181.75% annual)
 Sharpe: 0.0488



1.4 3. Results & Discussion

The model successfully generates 30-day price cones with adaptive confidence intervals. The **State Summary Table** provides a quick view of the dominant regime’s persistence over the past week. Key observations from the latest run (above):

- **Forecast Overlap Validation:** The model’s 7-day overlap period (dashed blue line) allows a direct comparison between forecasted paths and actual price movement. This serves as a sanity check for short-term model calibration.
- **Regime Probabilities:** When the maximum state probability exceeds 85%, the model exhibits higher conviction in its directional signal. Lower probabilities suggest market indecision.
- **Volatility Response:** The use of EWMA and a dynamic volatility floor helps the model adapt quickly to spikes in market turbulence (e.g., geopolitical events or regulatory news), as seen in the widened confidence bands.

1.4.1 Limitations

- The model assumes that the market regime will persist for the forecast horizon, which may not hold during structural breaks.
- HMM states are identified purely from price returns, ignoring on-chain metrics (e.g., hash rate, exchange flows) that could improve regime classification.

1.5 4. Conclusion

This project demonstrates a flexible framework for probabilistic Bitcoin price forecasting that combines **HMM-based regime detection** with an **enhanced GBM simulator**. By blending long-term state expectations with short-term market dynamics, the model provides a daily-updated, actionable signal for risk management and directional bias.

Future improvements could include: - Incorporating external macroeconomic indicators (e.g., DXY, S&P 500) as covariates in the HMM. - Replacing GBM with a stochastic volatility model (e.g., Heston) to better capture volatility smile. - Implementing an online learning mechanism to update HMM parameters incrementally.

The codebase is structured to allow easy integration with real-time data feeds, making it a viable component of a systematic trading infrastructure.

Note: The work described above on BTC forecasting is part of a larger multi-asset strategy framework. The following section outlines the architecture and progress of that broader system.

1.6 4. Extended Work: Multi-Asset Strategy Framework

This section provides context on the broader system within which the BTC forecasting model operates. While the preceding sections detail a standalone BTC prediction module, the Practicum project encompasses a larger, integrated multi-asset research platform.

1.6.1 4.1 System Overview

The full project aims to build an **automated strategy generation framework** that spans multiple asset classes (equities, fixed income, commodities, and cryptocurrencies). The core components are:

- **Rule Pool Mining + LightGBM Ensemble**

A library of thousands of simple technical rules (e.g., moving-average crossovers) is generated automatically. A LightGBM model learns to combine these weak rules into robust trading signals, originally validated on single assets (CSI 500, Bitcoin).

- **Macro Factor & News Sentiment Integration**

To address regime shifts triggered by monetary policy or geopolitical events, I incorporate external macro indicators and LLM-derived sentiment scores. This allows the system to adjust exposures when pure price-based signals are unreliable.

- **Dynamic Position Sizing**

Asset volatility varies dramatically (Bitcoin vs. Treasury bonds). A risk-budgeting module dynamically scales position sizes to maintain a target portfolio volatility, replacing static stop-loss rules.

- **C++ Parallelization**

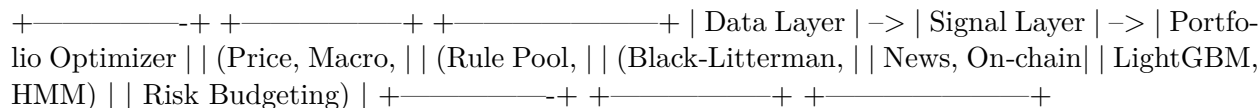
As the rule pool and asset universe grow, backtesting becomes computationally expensive. Selected modules have been re-implemented in C++ with multi-threading, achieving a ~40% reduction in runtime on an 8-core server. Further optimization using a task-based scheduler is underway.

- **Black-Litterman Multi-Asset Optimizer**

The most recent addition. It translates the system's asset-specific signals (e.g., "overweight

BTC”) and macroeconomic views (e.g., “inflation will surprise to the upside”) into optimal portfolio weights. The optimizer uses historical covariance matrices as a prior and blends in subjective views via the Black-Litterman framework.

1.6.2 4.2 Architecture Diagram (Conceptual)



1.6.3 4.3 Integration with the BTC Forecasting Module

The HMM+GBM model presented in Sections 2-3 serves two purposes within the broader system:

1. **Standalone signal for crypto exposure** – The daily “Trading Confidence” (LONG / NEUTRAL / SHORT) can be fed directly into the asset-allocation optimizer as a subjective view on Bitcoin’s expected return.
2. **Volatility forecasting** – The model’s output volatility `_daily` is used by the dynamic position sizer to calibrate the BTC allocation in real time.

1.6.4 4.4 Engineering Enhancements

To support the growing complexity of the platform, several “infrastructure” improvements have been made:

- **Unified Parameter Management**
All strategy parameters (stop-loss thresholds, target volatility, lookback windows) are now stored in a single YAML configuration file, reducing the risk of manual errors during experiments.
- **Exception Handling & Logging**
A robust error-capture mechanism prevents entire backtest runs from crashing due to isolated data anomalies, improving system reliability.

1.6.5 4.5 Business Value & Roadmap

The ultimate goal is to deliver a **production-ready multi-asset allocation system** that:

- Achieves a Sharpe ratio above **1.5** in backtests.
- Reduces maximum drawdown compared to single-asset strategies.
- Accelerates the research cycle (from overnight backtests to < 2 hours).

Next steps: - Complete the Black-Litterman optimizer integration and validate with historical data. - Implement a task-based parallel scheduler to further double backtesting speed. - Perform a small-scale live paper-trading test across equity, commodity, and crypto baskets.

1.7 5.ExFuture Work

1.7.1 5.1 BTC Forecasting Model

The HMM+GBM hybrid model provides daily probabilistic price cones and a directional confidence metric for Bitcoin. Backtests with 7-day overlap validation demonstrate reasonable adaptivity

to changing volatility regimes. The model can be integrated into a live trading pipeline as a volatility-aware signal generator.

1.7.2 5.2 Multi-Asset Strategy Framework

Beyond the BTC module, this Practicum project has advanced a comprehensive multi-asset research platform. Key accomplishments include: - A rule-mining + LightGBM signal layer that generalises across assets. - A Black-Litterman optimizer (under active development) to unify signals and macro views. - Engineering improvements that have already reduced backtesting time by ~40%.

The next phase will focus on completing the optimizer integration and validating the full system on historical data. The long-term objective is a robust, scalable platform capable of institutional-grade multi-asset allocation.

Future work may also explore: - Adding alternative data sources (e.g., on-chain metrics for crypto, satellite imagery for commodities). - Incorporating a reinforcement learning layer for adaptive parameter tuning.

1.8 5. References

1. Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*.
2. Hull, J. C. (2017). *Options, Futures, and Other Derivatives*. Pearson.
3. Binance API Documentation: <https://binance-docs.github.io/apidocs/>

1.9 Appendix: Output Files

The execution of the main script produces the following artifacts for each run: - `btc_forecast_track_YYYYMMDD.png` – Forecast cone plot. - `state_summary_YYYYMMDD.csv` – Last 7-day regime summary. - `forecast_result_YYYYMMDD.json` – Complete forecast percentiles and parameters. - `state_proba_history_YYYYMMDD.csv` – Time series of state probabilities.